

NYCU Introduction to Machine Learning, Homework 1

[Student ID], [Name]

The screenshot and the figures we provided below are just examples. **The results below are not guaranteed to be correct.** Please make sure your answers are clear and readable, or no points will be given.

Please also remember to convert it to a pdf file before submission.

You should use English to answer the questions (-5pt if not report in English).

After reading this paragraph, you should delete this paragraph.

Part. 1, Coding (60%):

(10%) Linear Regression Model - Closed-form Solution

1. (10%) Show the weights and intercepts of your linear model.

```
2024-03-14 23:13:57.944 | INFO | __main__:main:77 - LR_CF.weights=
```

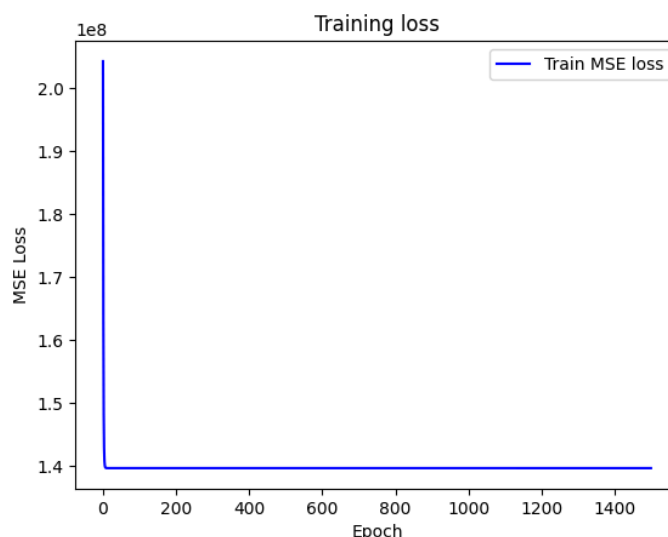
(40%) Linear Regression Model - Gradient Descent Solution

2. (10%)
 - Show the hyperparameters of your setting (e.g., learning rate, number of epochs, batch size, etc.).
 - Show the weights and intercepts of your linear model.

```
LR.gradient_descent_fit(train_x, train_y, lr=0.01, epochs=100)
```

```
2024-03-14 23:54:18.052 | INFO | __main__:main:84 - LR_GD.weights=
```

3. (10%) Plot the learning curve. (x-axis=epoch, y-axis=training loss)



4. (20%) Show your MSE.cf, MSE.gd, and error rate between your closed-form solution and the gradient descent solution.

```
2024-03-14 23:54:18.055 | INFO | __main__:main:93 - Prediction difference:
2024-03-14 23:54:18.055 | INFO | __main__:main:98 - mse_cf= , mse_gd= . Difference: 0.027%
```

(10%) Code Check and Verification

5. (10%) Lint the code and show the PyTest results.

```
● seanyu@bulbasaur:~/lectures/nycu/ml-and-pattern-recognition/hw1 *$ poetry run flake8 solution.py
○ seanyu@bulbasaur:~/lectures/nycu/ml-and-pattern-recognition/hw1 *$

===== test session starts =====
platform linux -- Python 3.9.5, pytest-8.0.2, pluggy-1.4.0
rootdir: /
collected 2 items

test_main.py 2024-03-16 11:52:21.189 | INFO | test_main:test_regression_cf:27 - model.weights=array([[3.]]), model.intercept=array([4.])
2024-03-16 11:52:21.198 | INFO | main:fit:57 - EPOCH 0, loss=3147.416663702691
2024-03-16 11:52:21.644 | INFO | main:fit:57 - EPOCH 10000, loss=0.29281584845965486
2024-03-16 11:52:22.094 | INFO | main:fit:57 - EPOCH 20000, loss=0.00536096424057785
2024-03-16 11:52:22.544 | INFO | main:fit:57 - EPOCH 30000, loss=9.815021195041223e-05
2024-03-16 11:52:22.998 | INFO | main:fit:57 - EPOCH 40000, loss=1.7969648133316264e-06
2024-03-16 11:52:23.450 | INFO | main:fit:57 - EPOCH 50000, loss=3.2899394472691304e-08
2024-03-16 11:52:23.905 | INFO | main:fit:57 - EPOCH 60000, loss=6.023324157052075e-10
2024-03-16 11:52:24.363 | INFO | test_main:test_regression_gd:39 - model.weights=array([3.]), model.intercept=3.9999966785390386
===== 2 passed in 4.37s =====
```

Part. 2, Questions (40%):

- (10%) Linear models $y = w^T x + b$ have limited fitting power.
 - In one sentence, explain why a single linear model is limited.
 - Give one concrete task that a single linear model cannot solve, and state why no single hyperplane/affine function solves it.

[Your answer here.]

- (15%) Why do we add a regularization term in linear regression? What are the differences between L2 regularization (Ridge) and L1 regularization (Lasso)? Please explain in detail.

[Your answer here.]

- (15%)
 - What is overfitting? Under what conditions can a model overfit? (List two) How can overfitting be alleviated? (List two)

[Your answer here.]